

COMPARATIVE ANALYSIS

1. Object Detection Performance Comparison

Two object detection models, Viola–Jones and YOLO, produce the following results: Model A has 80 True Positives, 20 False Positives, and 30 False Negatives, while Model B has 90 True Positives, 15 False Positives, and 20 False Negatives.

- (a) Calculate the precision and recall for both models.
- (b) Compare the results and determine which model provides better overall object detection performance. Justify your answer based on the calculated values.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce the following displacement errors in pixels: Method A: 2, 3, 2 and Method B: 3, 5, 4 for three consecutive frames.

- (a) Calculate the average displacement error for both methods.
- (b) Compare the two methods and determine which method provides more accurate motion estimation.
- (c) Justify your answer based on the error values.

3. Depth Estimation Comparison

A stereo vision system and a Structure-from-Motion (SfM) system estimate the depth of three objects. The estimated depths are: Stereo Vision: 2.0 m, 3.0 m, 4.0 m and SfM: 2.5 m, 3.8 m, 5.0 m, while the actual depths are 2 m, 3 m, and 4 m, respectively.

- (a) Calculate the absolute error for each object for both methods. (b) Calculate the average absolute error for both methods.
- (c) Compare the methods and identify which provides more accurate depth estimation.

1. Object Detection Performance Comparison

Two object detection models, **Viola–Jones (Model A)** and **YOLO (Model B)**, are evaluated using the following results:

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
Model A (Viola–Jones)	80	20	30
Model B (YOLO)	90	15	20

The performance of an object detection model is commonly measured using **Precision** and **Recall**. Precision indicates how many detected objects are actually correct, while Recall measures how many actual objects are successfully detected.

(a) Calculation of Precision and Recall

Formula

Precision

Recall

Model A (Viola–Jones)

Precision

$$= 80 / (80 + 20)$$

$$= 80 / 100$$

$$= \mathbf{0.80 = 80\%}$$

Recall

$$= 80 / (80 + 30)$$

$$= 80 / 110$$

$$= \mathbf{0.727 = 72.7\%}$$

Model B (YOLO)

Precision

$$= 90 / (90 + 15)$$

$$= 90 / 105$$

$$= \mathbf{0.857 = 85.7\%}$$

Recall

$$= 90 / (90 + 20)$$

$$= 90 / 110$$

$$= \mathbf{0.818 = 81.8\%}$$

(b) Comparison

Metric Model A Model B

Precision 80% **85.7%**

Recall 72.7% **81.8%**

The comparison shows that **YOLO performs better** in both precision and recall.

- Higher precision means YOLO produces fewer false detections.
- Higher recall means YOLO detects more actual objects and misses fewer targets.
- Viola–Jones is simpler and faster but has lower detection performance.

Advantages of YOLO

- Higher detection accuracy
- Detects multiple objects simultaneously
- Lower false positives
- Better performance in complex scenes

Advantages of Viola–Jones

- Fast processing
- Low computational requirement
- Suitable for basic face detection

Conclusion

Based on the calculated precision and recall values, **YOLO provides better overall object detection performance**. It has higher precision (85.7%) and higher recall (81.8%), making it more reliable for modern computer vision applications such as surveillance, autonomous vehicles, and smart traffic monitoring.

2. Motion Estimation Error Analysis

Two motion estimation methods produce displacement errors over three consecutive frames.

Frame Method A Method B

1	2	3
2	3	5
3	2	4

Motion estimation error indicates how accurately an algorithm estimates object movement between video frames. Lower error values indicate better performance.

(a) Average Displacement Error

Formula

Average Error

= (Sum of Errors) / Number of Frames

Method A

Average Error

= (2 + 3 + 2) / 3

= 7 / 3

= **2.33 pixels**

Method B

Average Error

= (3 + 5 + 4) / 3

= 12 / 3

= **4 pixels**

(b) Comparison

Method	Average Error
Method A	2.33 pixels
Method B	4 pixels

Method A has the lower average displacement error.

This means Method A estimates object motion more accurately than Method B.

(c) Justification

Method A produces consistently lower errors in every frame. Smaller displacement errors indicate that the estimated object position is closer to the actual position.

Method B produces larger errors, which may result in unstable tracking and inaccurate motion estimation.

Advantages of Method A

- Higher accuracy
- Stable tracking
- Better object localization
- Less sensitive to motion noise

Advantages of Method B

- May detect rapid movements
- Useful in highly dynamic scenes
- Suitable where small tracking errors are acceptable

Practical Example

In autonomous driving, accurate motion estimation is essential for predicting vehicle and pedestrian movement. Since Method A has a lower average error, it is more reliable for collision avoidance and path planning.

Conclusion

Method A is the better motion estimation technique because its average displacement error is only **2.33 pixels**, compared to **4 pixels** for Method B. Lower error means higher accuracy and more reliable object tracking.

3. Depth Estimation Comparison

A Stereo Vision system and a Structure-from-Motion (SfM) system estimate object depths.

Object Actual Depth Stereo Vision SfM

1	2 m	2.0 m	2.5 m
2	3 m	3.0 m	3.8 m
3	4 m	4.0 m	5.0 m

Depth estimation is important in robotics, autonomous vehicles, and 3D reconstruction because it helps determine the distance between the camera and objects.

(a) Absolute Error Calculation

Formula

Absolute Error

$$= |\text{Estimated Depth} - \text{Actual Depth}|$$

Stereo Vision

Object 1

$$|2.0 - 2.0| = 0 \text{ m}$$

Object 2

$$|3.0 - 3.0| = 0 \text{ m}$$

Object 3

$$|4.0 - 4.0| = 0 \text{ m}$$

SfM

Object 1

$$|2.5 - 2| = \mathbf{0.5\ m}$$

Object 2

$$|3.8 - 3| = \mathbf{0.8\ m}$$

Object 3

$$|5.0 - 4| = \mathbf{1.0\ m}$$

(b) Average Absolute Error

Stereo Vision

Average Error

$$= (0 + 0 + 0) / 3$$

$$= \mathbf{0\ m}$$

SfM

Average Error

$$= (0.5 + 0.8 + 1.0) / 3$$

$$= 2.3 / 3$$

$$= \mathbf{0.77\ m}$$

(c) Comparison

Method	Average Absolute Error
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Stereo Vision	0 m
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Structure-from-Motion	0.77 m
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Stereo Vision has zero error for all three objects, while Structure-from-Motion has noticeable errors.

Advantages of Stereo Vision

- Highly accurate depth estimation
- Reliable measurements
- Suitable for real-time applications
- Better for autonomous navigation

Advantages of Structure-from-Motion

- Does not require stereo cameras
- Can reconstruct 3D scenes from multiple images

- Useful for mapping and surveying

Practical Example

In autonomous vehicles, accurate depth estimation is necessary to determine the distance to nearby vehicles and pedestrians. Stereo Vision provides precise distance measurements, improving driving safety.

Structure-from-Motion is more suitable for creating 3D maps of buildings and landscapes where small depth errors are acceptable.

Conclusion

Stereo Vision provides the best depth estimation performance because its average absolute error is **0 m**, whereas Structure-from-Motion has an average error of **0.77 m**. Therefore, Stereo Vision is the preferred method for applications requiring accurate real-time distance measurement.